Jiangang Hao, Ph.D.

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Short Bio

- I am a research director at Educational Testing Service and leading the Assessment of Complex Skills program in the Research division. Curiosity is the fundamental driving force of all my research and work. My current research centers on leveraging AI and big data to understand people and their interactions, with particular focuses on behavioral data mining, game-based assessment, learning analytics, AI/machine learning, natural language processing, automated scoring, learning and assessment of 21st century skills, such as collaboration and communication.
- Besides scientific research itself, I also work extensively on developing and managing large-scale collaborative projects, from idea generation and planning to building teams to successful implementation. Solving real world problems using the most practical and effective methods is the core theme of all my research and work.
- Before joining in ETS, I worked on astrophysics and cosmology at Fermi National Accelerator Lab. I have been working on two prominent astronomical projects that embraced big data since early 2000s, the Sloan Digital Sky Survey (SDSS) and the Dark Energy Survey (DES). My research ranged from developing CCD camera and astronomical digital images processing pipeline to developing algorithms for galaxy clustering and cosmological parameter estimation.

Education

- **University of Michigan, Ann Arbor, USA**
  PhD, Physics (2009); MA, Statistics (2006)
- **East China University of Science and Technology, Shanghai, China**
  DSc, Applied Mathematics (2004); MS, Physics (2001); BS, Physics (1998)

Employment

- **Educational Testing Service, Princeton, NJ, USA**
  - *Director*, Assessment of Complex Skills Program, 9/2023 - now
  - *Director*, Psychometric and Data Science Modeling, 10/2020 - 9/2023
  - *Managing Senior Research Scientist*, Assessment Analytics and Data Science, 3/2020 - 10/2020
  - *Senior Research Scientist*, Psychometrics, Statistics and Data Science, 8/2016 - 3/2020
  - *Research Scientist*, Center for Computational Psychometrics, 5/2013 - 8/2016
- **Fermi National Accelerator Laboratory, Batavia, IL, USA**
  - *Visitor*, Center for Particle Astrophysics, 5/2013 - 2019
  - *Postdoc*, Center for Particle Astrophysics, 5/2009 - 5/2013

Employment (continued)

- **University of Pennsylvania, Philadelphia, PA, USA**
  - *Visiting Scientist*, Department of Physics, 1/2012 - 5/2013
- **University of Michigan, Ann Arbor, MI, USA**
  - *Research Assistant*, Department of Physics, 6/2006 - 5/2009
  - *Graduate Student Instructor*, Department of Physics, 9/2004 - 6/2006

Corporate Services and Leadership

- Member, Enterprise test security leadership team, Educational Testing Service, 2022 - now.
  Key responsibility: oversee the development and implementation of the test security initiative
- Member, Enterprise test security strategy core team, Educational Testing Service, 2022 - now.
  Key responsibility: plan and coordinate the implementation of research efforts on data-driven test security
- Lead, Initiative for training in data science, AI and digital learning, Educational Testing Service, 2021 - 2022.
  Key responsibility: lead the initiative and coordinate the development and implementation of upskill training courses on AI, data science and digital learning
- Member, R&D tech steering committee, Educational Testing Service, 2019 - now.
  Key responsibility: oversee the R&D computing infrastructure and cloud computing and provide technical guidance.
- Co-chair, Research Advisory Council, Division of Psychometrics, Statistics and Data Science (PSDS), Educational Testing Service, 2019 - 2020.
  Key responsibility: develop the measurement frontier research agenda for PSDS division, review and select research proposals, and oversee the implementation of the research projects
- Member, ETS AI strategy committee, Educational Testing Service, 2019 - 2020.
  Key responsibility: develop ETS corporate AI strategy
- Member, Remote proctoring strategy committee, Educational Testing Service, 2020 - 2021.
  Key responsibility: Develop ETS' remote proctoring strategy and plan implementation
- Lead, Computational Psychometrics sub-initiative, Educational Testing Service, 2016 - 2017.
  Key responsibility: develop research agenda around computational psychometrics and oversee the implementation of approved research projects
- Co-lead, Software and Data Infrastructure sub-initiative, Educational Testing Service, 2014 - 2016.
  Key responsibility: develop research agenda around software and data infrastructure and oversee the implementation

Technical Expertise

- Data Science, Machine Learning/AI, Natural Language Processing
- Statistics, Psychometrics, Scientific Computing
- Game/simulation-based Assessments, Assessment of 21st Century Skills and interpersonal skills
- Dynamical System and Differential Equations
- Image Segmentation, Registration, Pattern Recognition
- Charge Coupling Device (CCD) for Astronomical Imaging

Technical Expertise (continued)

- Astrophysics, Cosmology, General Relativity

Programming Skills

- Python/R (see my github at: https://github.com/jgbrainstorm)
- C++ (see my package at: https://sites.google.com/site/jiangangecgmm)
- Cloud Computing (Amazon Web Service)

Patents

- Deane, P., Choi, Ikkyu, **Hao, J.**, and Zhang, M. Educational Testing Service (2023). Behavior-based electronic essay assessment fraud detection. U.S. Patent 11,630,896. Washington, DC: U.S. Patent and Trademark Office.
- Zhang, M., **Hao, J.**, Deane, P. and Li, C., Educational Testing Service (2021). Defining personalized writing burst measures of translation using keystroke logs. U.S. Patent 11,195,431. Washington, DC: U.S. Patent and Trademark Office.
- Zhang, M., **Hao, J.** and Deane, P., Educational Testing Service (2021). Generating scores and feedback for writing assessment and instruction using electronic process logs. U.S. Patent 10,964,224. Washington, DC: U.S. Patent and Trademark Office.
- Deane, P., Feng, G., Zhang, M., **Hao, J.**, Bergner, Y., Flor, M., ...and Lederer, N. (2019). Generating Scores and Feedback for Writing Assessment and Instruction Using Electronic Process Logs, U.S. Patent No. 14/937,164, Washington, DC: U.S. Patent and Trademark Office.

Books

- *Computational Psychometrics: New Methodologies for a New Generation of Digital Learning and Assessment*, Alina A. von Davier, Robert J. Mislevy, and Jiangang Hao (Eds). Springer, 2021
- *The Meaning of Relativity*, by Albert Einstein. Chinese translation by Jiangang Hao, Daojun Liu and Xinzhou Li. Shanghai Science and Technological Education Press, 2001

Media Reports

- ETS News, 2019, *Got a problem? here is a 21st century way to solve it*
- PRnewswire, 2014, *ETS Researchers Connect Games and Assessment in Manual for Designers*
- Fermilab Today, 2011, *Fermilab postdoc improves Google Sky galaxy views*
- Wired, 2010, *Google Sky Adds Galaxy Clusters*
- MIT Technology Review, 2010, *Galaxy Clusters in Google Sky*

Mentored Student/Interns

- **Kehinde Elelu**, 2024 ETS Summer Intern (Civil Engineering Ph.D. Student of Clemson University)
- **Yang Zhong**, 2023 ETS Summer Intern (Computer Science Ph.D. Student of University of Pittsburgh)
- **Daniel Adams**, 2018 ETS Summer Intern (Psychometrics Ph.D. Student of University of Wisconsin - Madison), now ETS Psychometrician

Mentored Student/Interns (continued)

- **Yang Jiang**, 2017 ETS Summer Intern (Cognitive Science Ph.D. Student of Columbia University), now ETS Research Scientist
- **Lu Ou**, 2016 ETS Summer Intern (Statistics Ph.D. Student of Penn State Univeristy), Now at Netflix
- **Thales Ricarte**, 2015 ETS Exchange Student (Psychometrics Ph.D. student of University of São Paulo)
- **Zuowei Wang**, 2014 ETS Summer Intern (Psychology Ph.D. student of University of Michigan), Now ETS Research Scientist

Awards and Recognition

- *Annual Award for Exceptional Achievement in Educational Measurement*, 2024, National Council on Measurement in Education (NCME), USA
- *Presidential Awards*, 2016, Educational Testing Service, USA
- *SPOT Awards*,2013-2015, Educational Testing Service, USA
- *Most Valued Reviewer*, 2011, Astroparticle Physics, Elseiver
- *Builder*, 2011, Dark Energy Survey Collaboration
- *GROCS Award*,2009, University of Michigan, USA
- *First Prize for the Advancement of Science and Technology in Shanghai*, 2005, Shanghai, China

Professional References

- **Dr. Alina von Davier**, Chief of Assessment, Duolingo Inc. Email: avondavier@duolingo.com
- **Dr. Robert Mislevy**, Lord Chair Emeritus, Educational Testing Service, Email: rmislevy@umd.edu
- **Dr. Tinkam Ho**, Senior AI Scientist, IBM, Email: tho@us.ibm.com
- **Prof. Timothy McKay**, Professor of Physics, Astronomy and Education, University of Michigan, Email: tamckay@umich.edu

Academic Services

Editorship and Reviewership

- *Associate editor*, Frontiers in Psychology: Quantitative Psychology and Measurement
- *Reviewer* of the following educational research journals (active)
  - *Psychometrika*
  - *Journal of Educational Measurement*
  - *Educational and Psychological Measurement*
  - *Journal of Educational and Behavioral Statistics*
  - *International Journal of Testing*
  - *Learning and Instruction*
  - *Computers and Education*
  - *PLOS One*
  - *Journal of Learning Analytics*
  - *Computers in Human Behavior*
  - *Journal of Educational Data Mining*
  - *Educational Assessment*
  - *IEEE Transactions on Learning Technology*
  - *Quality Assurance in Education*
  - *ETS Research Report*
- *Reviewer* of the following physics research journals (inactive)
  - *Astrophysical Journal Supplement Series* (Impact Factor: 13.456)
  - *Astrophysical Journal* (Impact Factor: 6.024)
  - *Physics Letters B* (Impact Factor: 3.955)
  - *Astroparticle Physics* (Impact Factor: 3.216)

Academic Services (continued)

Conference Program Committee

- 2024, The 16th International Conference on Computer Supported Education, Angers, France
- 2024, The 25th International Conference on Artificial Intelligence in Education, Recife, Brazil
- 2024, The 17th International Conference on Educational Data Mining, Atlanta, Georgia, USA
- 2023, The 37th AAAI conference on artificial intelligence, Washington DC, USA
- 2023, The 16th International Conference on Educational Data Mining, Bengaalura, India
- 2023, The 18th workshop on innovative use of NLP for building Educational Applications, Seattle, USA
- 2023, The 3rd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 12th International Joint Conference on Natural Language Processing
- 2022, The 36th AAAI conference on artificial intelligence, virtual
- 2022, The 15th International Conference on Educational Data Mining, Durham, UK
- 2022, The 17th workshop on innovative use of NLP for building Educational Applications, Seattle, USA
- 2022, The 2nd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 12th International Joint Conference on Natural Language Processing
- 2021, The 35th AAAI conference on artificial intelligence, virtual
- 2021, The 14th International Conference on Educational Data Mining, Paris, France
- 2020, The 15th workshop on innovative use of NLP for building Educational Applications, Seattle, USA
- 2020, The 13rd International Conference on Educational Data Mining, Ifrane, Morocco (changed to virtual)
- 2019, The 12nd International Conference on Educational Data mining, Montreal, Canada
- 2019, The 57th Annual Meeting of the Association for Computational Linguistics (ACL), Florence, Italy
- 2019, The 14th workshop on innovative use of NLP for building Educational Applications, Florence, Italy
- 2018, The 11th International Conference on Educational Data mining, Buffalo, NY, USA
- 2018, The 13rd Workshop on Innovative Use of NLP for Building Educational Applications, New Orleans, Louisiana, USA
- 2018, The 16th Annual Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, New Orleans, Louisiana, USA
- 2018, The 13th International Conference on Learning Sciences, London, UK
- 2017, The 10th International Conference on Educational Data mining, Wuhan, China
- 2017, The 12th Workshop on Innovative Use of NLP for Building Educational Applications, Copenhagen, Denmark
- 2015, IEEE International conference on systems, man and cybernetics, Hong Kong, China

Academic Services (continued)

Workshop/Symposium Chair/Co-Chair

- Coordinated session "Support Test Security of Remote Testing with Process Data Analytics and AI", 2023 NCME, Chicago, IL
- Training workshop "Addressing the Data Challenges of Next-generation Assessments: Data Science Upskilling for Psychometricians", 2023 NCME, Chicago, IL
- Coordianted session "Assessing Collaborative Problem Solving at Scale: Individual Contribution to Teamwork", 2023 NCME, Chicago, IL
- Panel discussion "From Science Fiction to 2023's GPT-4: Implications for Assessment", 2023 NCME, Chicago, IL
- Symposium "AI and data analytics approaches to address test security issues from remotely proctored tests", 2022 Conference on Test Security, Princeton, NJ, Nov. 2022
- Training workshop "Machine Learning, Natural Language Processing, and their Application in Educational Assessment, 2022 MARC conference, Nov. (2022).
- Pre-conference Workshop "Addressing the psychometrics challenges from digital learning and assessment: how computational psychometrics helps", Edtech & Computational Psychometrics Summit, June (2022)
- Training workshop "Addressing the data challenges from next-generation assessments: data science upskilling for psychometricians", NCME, Virtual (2022)
- Coordinated session "Using Response Process Data to Support At-home Administrations of High-stakes Assessments", NCME, Virtual (2022)
- Training workshop "Addressing the data challenges from next-generation assessments: data science upskilling for psychometricians", NCME, Virtual (2021)
- Coordinated session "Leveraging Response Process Data to Support Testing Programs: Strategies and Real-world Examples", NCME, Virtual (2021)
- Coordinated session, "Assessing collaborative problem solving at scale: the status quo and the next", NCME, Virtual (2020)
- Workshop "EPCAL: studying synchronous collaboration at scale", 13th International Conference on Computer Supported Collaborative Learning, Lyon, France (2019)
- Workshop "Evidence centered design and computational psychometric solution for process data", annual meeting of National Council on Measurement in Education, New York, NY, USA (2018)
- Workshop "EPCAL: computer-supported collaboration at scale", 12th International Conference on Computer Supported Collaborative Learning, Philadelphia, PA, USA (2017)
- Workshop "Evidence centered design and computational psychometric solution for process data", annual meeting of National Council on Measurement in Education, San Antonio, TX, USA (2017)

Key Research Projects

- **Projects around the assessment of collaborative problem sovling and 21st century skills** [2013 - now]
  Leading/co-leading a sets of research projects around assessments of collaborative problem solving and 21st century skills since 2013. The details are as follows:

  - *Collaborative Science Assessment Prototype (ECSAP)* [2013 - 2017]. Leading this project to develop assessment prototype of collaborative problem solving in the domain of science. The prototype has been selected by Peggy Carr (NCES commissioner) as an example of future assessments in her presentation at the White House in 2014. Some findings from the project can be found in this recent paper: https://onlinelibrary.wiley.com/doi/full/10.1002/ets2.12276

  - *ETS Platform for Collaborative Assessment and Learning (EPCAL)* [2015 - now]. Leading this project to develop online platform for assessment and learning of human-human interactions, e.g., collaboration, communication, and social emotional learning. It is a premium platform for conducting research on human-human interaction, and is supporting over 20 projects over the years funded by ETS and external funding agencies such as NSF, IES, ARI, Gates Foundation, and others. A description of the platform can be found here: https://onlinelibrary.wiley.com/doi/full/10.1002/ets2.12181

  - *ETS Platform for Collaborative Human Annotation (EPCHA)* [2020 - now]. Leading this project to develop a distributed platform that simplify the human coding of the communication data from collaborative tasks.

  - *Collaborative Problem Solving Skills: Estimating an Individual's Contribution to Small Group Performance* [2019 - 2022]. Co-PI of this project funded by the Army Research Institute to investigate how to measure individual's contribution to teamwork.

  - *Improve learning and assessments through human-human interaction*[2020 - now]. Leading this multi-year project funded by ETS research allocation to explore ways of improving learning and assessment by using evidence from human-human interactions. A paper summarize the psychometric considerations and general scoring strategies for assessing collaborative problem solving: https://onlinelibrary.wiley.com/doi/full/10.1002/ets2.12276.

  - *Developing Middle Grade Students Social Emotional Learning Skill through Technology Enhanced Collaborative Learning* [2024-2028]. I am the co-PI of this project funded by the Education Innovation Research initiative of the Department of Education. See this link: https://www.ed.gov/news/press-releases/biden-harris-administration-announces-277-million-education-innovation-and-research-grants-help-address-academic-recovery

Key Research Projects (continued)

- **Projects around data analytics and AI to support test security** [2020 - now]
  Since 2020, I have been leading multiple data-driven projects funded by the Research Allocation and ETS test security initiative to innovate analytics and AI methods to support test security, especially for remotely administered tests. These projects have yielded several significant products, contributing to over 60% of cheating detection. Listed below are some of the tools we developed:

  - *AutoESD - Automated Essay Similarity Detection: A system that identifies similar essays, presenting them to human experts for final decisions via an AI-assisted analytics dashboard.*
  - *AutoSSD - Automated Speech Similarity Detection is a system that detects similar speech responses and supply them through an analytics and AI assisted dashboard for human experts to make final decisions.*
  - *AI-Reviewer: AI-based essay/speech similarity reviewer is a machine learning system that utilizes a number of similarity features from essay/speech responses and human expert ajudications to make decisions automatically.*
  - *Keystroke analytics and biometrics include a set of tools that are used to identify imposters, copy writing, and AI generated essays.*
  - *Clickstream data-based Suspicious behavioral detection includes a set of tools that identify anormly behaviors in test taking process.*
  - *Detection of AI (ChatGPT, GPT-3) generated essays includes a set of tools to detect AI generated essays through machine learning, writing process and essay similarity.*

- **Projects around the development of core data and software infrastructure for game/simulation-based (virtual performance) assessments** [2014 - 2021]. I've been spear-heading a series of projects dedicated to developing data analytics solutions for telemetry logs from game-based assessments and digital learning environments. Listed below are some key projects:

  - *GlassPy: game log analysis in python* [2014 - 2019]. I led this project to develop python package to address the analysis of log data analysis from games. Here is an early paper from the project https://onlinelibrary.wiley.com/doi/full/10.1002/ets2.12096. A more recent presentation could be found at: https://youtu.be/pcWXjS830fo
  - *Data model for VPAs and Evidence Trace File* [2014 - 2018]. I led this project to develop the ETS data model for virtual performance assessment and we proposed the evidence trace file to replace the log file. We also developed the evidence identification centered data design (EICDD) as an interface between ECD and data engineering. Here is paper about the details of evidence trace file and our data model: https://onlinelibrary.wiley.com/doi/full/10.1002/ets2.12215.
  - *EPCAL analytics* [2019 - 2021]. I led this project to develop analytics dashboards for the data from collaborative tasks deployed on the EPCAL platform.

Key Research Projects (continued)

- **Projects around the application of data science, machine learning/AI and Natural Language Processing to improve learning and assessment** [2013 - now]. I have been leading, co-leading, and participating in a variety of projects aimed at enhancing learning and assessment methods through the utilization of response process data. These projects encompass a range of innovative approaches, as detailed below:
  - CPS-rater: This project focuses on the automated annotation of communication in collaborative activities, streamlining the analysis of interactions in group settings.
  - Writing Analytics and Biometrics: Here, the emphasis is on leveraging keystroke data during the writing process. This approach allows for a nuanced understanding of writing habits and skills.
  - Learner Behavioral Analytics in Writing Mentor: This initiative analyzes user interaction with the online writing application, Writing Mentor, to gain insights into learning behaviors and patterns.
  - Winsight Process Data Analytics: Contributing to the field of psychometrics, this project employs process data analytics to support and enhance psychometric research and practices.
  - Analysis and modeling of Process Data from the National Assessment of Educational Progress (NAEP): In this project, process data from NAEP is scrutinized to derive deeper insights into educational progress and trends.

- **Dark Energy Survey** [2009 - 2013]. I worked on the dark energy survey project while I was with Fermi National Accelerator Lab.
  - DECam focal plane flatness measurement: I developed a optical method to measure the flatness of the focal plane of DECam.
  - CCD image recuction and image quality pipeline: I developed the image reduction and image quality measurement pipeline for testing CCDs for DECam.
  - Galaxy clustering algorithm development: as a continuation of my PhD project, I completed the GMBCG algorithm for galaxy cluster detection.
  - Galaxy alignment measurement: I performed the precision measurement of galaxy alignment in clusters based on SDSS data.

- **Sloan Digital Sky Survey** [2004 - 2009]. During my PhD, my research centers on developing unsupervised machine learning algorithm to detect cluster galaxies based on data from the Sloan Digital Sky Survey.
  - Error corrected Gaussian Mixture Model: I developed an extended gaussian mixture model that accounts for the measurement errors. Used this algorithm, we precisely measured the red sequence ridgeline evolution at different redshit.
  - GMBCG algorithm: I developed the Gaussian Mixture Bright Central Galaxy (GMBCG) algorithm and created the largest catalog of optical galaxy clusters as of 2013.

Publications - click to see the Google Scholar page

1 **Hao, J.**, von Davier, A. A., Yaneva, V., Lottridge, S., von Davier, M., & Harris, D. J. (2024). Transforming assessment: The impacts and implications of large language models and generative ai. *Educational Measurement: Issues and Practice*, *43*(2), 16–29.

2 Jiang, Y., **Hao, J.**, Fauss, M., & Li, C. (2024). Detecting chatgpt-generated essays in a large-scale writing assessment: Is there a bias against non-native english speakers? *Computers & Education*, *217*, 105070.

3 **Hao, J.**, & Fauss, M. (2023). Test security in remote testing age: Perspectives from process data analytics and ai. *In H. Jiao and R.W. Lissitz (Eds.), Machine learning, natural language processing and psychometrics*.

4 Jiang, Y., Martin-Raugh, M., Yang, Z., **Hao, J.**, Liu, L., & Kyllonen, P. C. (2023). Do you know your partner's personality through virtual collaboration or negotiation investigating perceptions of personality and their impacts on performance. *Computers in Human Behavior*, *141*, 107608.

5 Yan, D., Fauss, M., **Hao, J.**, & Cui, W. (2023). Detection of ai-generated essays in writing assessment. *Psychological Testing and Assessment Modeling*, *65*(2), 125–144.

6 Zu, J., Choi, I., & **Hao, J.** (2023). Automated distractor generation for fill-in-the-blank items using a prompt-based learning approach. *Psychological Testing and Assessment Modeling*, *65*(2), 55–75.

7 Castellano, K. E., Sinharay, S., **Hao, J.**, & Li, C. (2022). An investigation into the impact of test session disruptions for at-home test administrations. *Applied Psychological Measurement*, 01466216221128011.

8 Choi, I., **Hao, J.**, Deane, P., & Zhang, M. (2021). Benchmark keystroke biometrics accuracy from high-stakes writing tasks. *ETS Research Report Series*.

9 Flor, M., & **Hao, J.** (2021). Text mining and automated scoring. In *Computational psychometrics: New methodologies for a new generation of digital learning and assessment* (pp. 245–262). Springer.

10 **Hao, J.** (2021). Supervised machine learning. In *Computational psychometrics: New methodologies for a new generation of digital learning and assessment* (pp. 159–171). Springer.

11 **Hao, J.**, & Mislevy, R. J. (2021). A data science perspective on computational psychometrics. In *Computational psychometrics: New methodologies for a new generation of digital learning and assessment* (pp. 133–158). Springer.

12 Rosen, Y., Stoeffler, K., Simmering, V., **Hao, J.**, & Davier, A. v. (2021). Development of scalable assessment for collaborative problem-solving. In *International handbook of computer-supported collaborative learning* (pp. 517–532). Springer.

13 von Davier, A. A., Mislevy, R. J., & **Hao, J.** (2021). Introduction to computational psychometrics: Towards a principled integration of data science and machine learning techniques into psychometrics. In *Computational psychometrics: New methodologies for a new generation of digital learning and assessment* (pp. 1–6). Springer.

14 Martin-Raugh, M. P., Kyllonen, P. C., **Hao, J.**, Bacall, A., Becker, D., Kurzum, C., … Barnwell, P. (2020). Negotiation as an interpersonal skill: Generalizability of negotiation outcomes and tactics across contexts at the individual and collective levels. *Computers in Human Behavior*, *104*, 105966.

15 **Hao, J.**, & He, Q. (2019). Assessment practice of collaborative problem solving. *China Exam*, *9*.

16 **Hao, J.**, & Ho, T. K. (2019). Machine learning made easy: A review of scikit-learn package in python programming language. *Journal of Educational and Behavioral Statistics*, *44*(3), 348–361.

17 **Hao, J.**, Liu, L., Kyllonen, P., Flor, M., & von Davier, A. A. (2019). Psychometric considerations and a general scoring strategy for assessments of collaborative problem solving. *ETS Research Report Series*.

18 **Hao, J.**, & Mislevy, R. J. (2019). Characterizing interactive communications in computer-supported collaborative problem-solving tasks: A conditional transition profile approach. *Frontiers in psychology*, *10*, 1011.

19 **Hao, J.**, Zapata-Rivera, D., Andrews-Todd, J., Zhu, M., & Mitros, P. (2019). Epcal: Studying synchronous online collaboration at scale. *A Wide Lens: Combining Embodied, Enactive, Extended, and Embedded Learning in Collaborative Settings*.

20 Lee, Y.-H., **Hao**, **J.**, Man, K., & Ou, L. (2019). How do test takers interact with simulation-based tasks? a response-time perspective. *Frontiers in psychology*, *10*, 906.

21 Stadler, M., Fischer, F., Graesser, A., Papadopoulos, P. M., Radkowitsch, A., Schmidmaier, R., … Greiff, S. (2019). Ghost in the machine: A symposium on collaboration between human and computerized agents in educational contexts.

22 **Hao**, **J.**, & Mislevy, R. J. (2018). The evidence trace file: A data structure for virtual performance assessments informed by data analytics and evidence-centered design. *ETS Research Report Series*, *2018*(1), 1–16.

23 **Hao**, **J.**, Zapata-Rivera, D., Graesser, A. C., Cai, Z., Hu, X., & Goldberg, B. (2018). –towards an intelligent tutor for teamwork: Responding to human sentiments. *Design recommendations for intelligent tutoring systems*, *6*, 151–160.

24 Zhang, M., **Hao**, **J.**, Li, C., & Deane, P. (2018). Defining personalized writing burst measures of translation using keystroke logs. In *Proceedings of the 2018 educational data mining conference* (pp. 549–552).

25 Andrews, J. J., Kerr, D., Mislevy, R. J., von Davier, A., **Hao**, **J.**, & Liu, L. (2017). Modeling collaborative interaction patterns in a simulation-based task. *Journal of Educational Measurement*, *54*(1), 54–69.

26 Dingler, C., von Davier, A. A., & **Hao**, **J.** (2017). Methodological challenges in measuring collaborative problem-solving skills over time. *Team dynamics over time*, 51–70.

27 Halpin, P. F., von Davier, A. A., **Hao**, **J.**, & Liu, L. (2017). Measuring student engagement during collaboration. *Journal of Educational Measurement*, *54*(1), 70–84.

28 **Hao**, **J.**, Chen, L., Flor, M., Liu, L., & von Davier, A. A. (2017). Cps-rater: Automated sequential annotation for conversations in collaborative problem-solving activities. *ETS Research Report Series*, *2017*(1), 1–9.

29 **Hao**, **J.**, Liu, L., von Davier, A. A., & Kyllonen, P. C. (2017). Initial steps towards a standardized assessment for collaborative problem solving (cps): Practical challenges and strategies. In *Innovative assessment of collaboration* (pp. 135–156). Springer, Cham.

30 **Hao**, **J.**, Liu, L., von Davier, A. A., Lederer, N., Zapata-Rivera, D., Jakl, P., & Bakkenson, M. (2017). Epcal: Ets platform for collaborative assessment and learning. *ETS Research Report Series*, *2017*(1), 1–14.

31 Liu, L., **Hao**, **J.**, Andrews, J. J., Zhu, M., Mislevy, R. J., Kyllonen, P., … Graesser, A. (2017). Collaborative problem solving: Innovating standardized assessment. Philadelphia, PA: International Society of the Learning Sciences.

32 Shu, Z., Bergner, Y., Zhu, M., **Hao**, **J.**, & von Davier, A. A. (2017). An item response theory analysis of problem-solving processes in scenario-based tasks. *Psychological Test and Assessment Modeling*, *59*(1), 109.

33 von Davier, A. A., **Hao**, **J.**, Liu, L., & Kyllonen, P. (2017). Interdisciplinary research agenda in support of assessment of collaborative problem solving: Lessons learned from developing a collaborative science assessment prototype. *Computers in Human Behavior*, *76*, 631–640.

34 Zapata-Rivera, D., Liu, L., Chen, L., **Hao**, **J.**, & von Davier, A. A. (2017). Assessing science inquiry skills in an immersive, conversation-based scenario. In *Big data and learning analytics in higher education* (pp. 237–252). Springer, Cham.

35 Flor, M., Yoon, S.-Y., **Hao**, **J.**, Liu, L., & von Davier, A. (2016). Automated classification of collaborative problem solving interactions in simulated science tasks. In *Proceedings of the 11th workshop on innovative use of nlp for building educational applications* (pp. 31–41).

36 **Hao**, **J.**, Liu, L., von Davier, A. A., Kyllonen, P., & Kitchen, C. (2016). Collaborative problem solving skills versus collaboration outcomes: Findings from statistical analysis and data mining. In *Proceedings of the 9th international conference on educational data mining*.

37 **Hao, J.**, Smith, L., Mislevy, R., von Davier, A., & Bauer, M. (2016). Taming log files from game/simulation-based assessments: Data models and data analysis tools. *ETS Research Report Series*, *2016*(1), 1–17.

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Presentations

1. **Hao, J.**, (April 2024), Explorations of GPT, Discussant, NCME 2024, Philadelphia, PA.
2. **Hao, J.**, (April 2024), Characterizing individual collaborative problem solving through conditional transistion profle: stability across tasks and teams. NCME 2024, Philadelphia, PA.
3. Zhong, Y., **Hao, J.**, and Fauss, M. (April 2024), Comparison of essays generated by different LLMS, Discussant, NCME 2024, Philadelphia, PA.
4. **Hao, J.**, (April 2024), One year after GPT-4: how AI changed Assessments, Organized panel discussion at NCME 2024, Philadelphia, PA.
5. **Hao, J.**, and Fauss, M. (April 2024), Practical considerations of using detectors of AI-generated essays: dos and don't, NCME 2024, Philadelphia, PA.
6. Fauss, M., and **Hao, J.**, (April 2024), Effects of sampling temperature on writing style and quality, NCME 2024, Philadelphia, PA.
7. Jiang, Y., **Hao, J.**, Fauss, M., and Li, C. (April 2024), Towards investigating the fairness of detecting LLM-generated essays, NCME 2024, Philadelphia, PA.
8. **Hao, J.**, Fauss, M., and Li, C. (April 2024), Detecting AI-generated essays: what you should know before making important decisions, AERA 2024, Philadelphia, PA.
9. Jiang, Y., **Hao, J.**, Fauss, M. and Li, C. (April 2024), Detecting AI-generated essays in a large-scale educational assessment: is there bias against non-native English speakers?, AERA 2024, Philadelphia, PA.
10. **Hao, J.**, (Nov. 2023), Assessing collaborative problem solving with interactive digital technology and AI, Presentation to the ETS Visiting Panel for Research.

11. **Hao, J.**, (June 2023), Generative AI: Challenges and opportunities for assessment, ATP-ECPS 2023
12. **Hao, J.**, (July 2023), ChatGPT and advanced generative AI: Impact on data analysis and data science, Presentation at ETS DA seminar series.
13. **Hao, J.**, (July 2023), Assessing collaborative problem solving at scale – psychometric challenges and strategies, IMPS 2023, College Park, MD.
14. **Hao, J.**, (June 2023), Assessing interpersonal skills at scale, Presentation given at ETS summer internship seminar.
15. **Hao, J.**, (July 2023), Generative AI: Challenges and opportunities for assessment, Presentation given at ETS summer internship seminar.
16. **Hao, J.**, (April 2023), ChatGPT, GPT4+ and the Future of Assessments, NCME 2023, Chicago, IL.
17. **Hao, J.**, Kerzabi, E., and Kyllonen, P., (April 2023), Assessing CPS at Scale – EPCAL ecosystem and psychometric considerations, NCME 2023, Chicago, IL
18. **Hao, J.**, (Jan 2023), Detection of ChatGPT generated essays in TOEFL, ETS Comopany-wide presentation, Princeton, NJ
19. Fauss, M., and **Hao, J.**, (Jan. 2023), Automated Speech Similarity Detection (AutoSSD) in TOEFL iBT Speaking, ETS Company-wide presnetation, Princeton, NJ
20. **Hao, J**. and Li, C., (Nov. 2022), Detecting remote desktop access using AI and clickstream process data, 2022 Conference on Test Security, Princeton, NJ, Nov. 2022
21. Choi, I., **Hao, J.**, Deane, P., and Zhang, Mo., (Nov. 2022), Benchmark Keystroke Biometrics Accuracy from High-Stakes Writing Tasks, 2022 Conference on Test Security, Princeton, NJ, Nov. 2022
22. Zhang, M., Deane, P., and **Hao, J.**, (Nov. 2022), Detection of Retyping vs. Drafting Through AI-based Methods based on Keystroke Process Data, 2022 Conference on Test Security, Princeton, NJ, Nov. 2022
23. Novak, J., Choi, I., **Hao, J.**, and Li, C., (Oct. 2022), AutoESD: An Automated Solution to Detect Copied Essays, 2022 Conference on Test Security, Princeton, NJ, Oct. 2022
24. Yan, D., Fauss, M., Cui, W., and **Hao, J.**, (Oct. 2022), Detection of AI-generated Essays, 2022 Conference on Test Security, Princeton, NJ, Oct. 2022
25. Deane, P., Zhang, M., and **Hao, J.**, (Oct. 2022), Detecting Writing-Process Characteristics Associated with Non-Genuine Text Generation, 2022 Conference on Test Security, Princeton, NJ, Oct. 2022
26. **Hao, J**., (Nov. 2022), Using Data Analytics and AI to Support Test Security of Remotely Proctored Tests, Invited presentation at 2022 MARC conference.
27. **Hao, J**., (July 2022), Response Process Data from Digital Assessments: Challenges and Strategies, Presentation given to ETS summer interns
28. **Hao, J**., (June 2022), Streamlit Dashboard and AWS Deployment: Unleashing the Power of Interactive Reports, Presentation given at ETS DA Seminar.
29. Liu, L., and **Hao, J**., (April 2022), Computer-Supported Collaborative Modeling Activities to Promote Science Learning, presentation given at AERA 2022
30. **Hao, J**., (March 2022), RMS research on test security, Presentation given to SBU at ETS.
31. Novak, J., Choi, I, **Hao, J.**, and Li, (April 2022), AutoESD: An Automated Solution to Detect Copied Essays, Presentation given to ETS employee.
32. **Hao, J**., (Sept. 2021), Using clickstream process data to support remotely proctored high-stakes tests: a synergy between psychometrics, data science and AI, Presentation given to ETS employee.

33. Zhu, M., Burstein, J., **Hao, J**. (June, 2021), Uncovering patterns of use in the writing mentor app through cluster analysis, Presentation given at NCME 2021

34. **Hao, J**., (July 2021), Psychometric and data science modeling at ETS, Presentation given to 2021 ETS summer interns.

35. **Hao, J**., (April 2021), Beyond a single score: scoring and reporting strategy for scalable assessments of collaborative problem solving, Presentation given at 2021 conference of International Testing Commission (ITC).

36. **Hao, J.**, (Dec. 4th, 2020), Characterizing the quality of simulation-based task: edit distance-based measures, Presentation given at the virtual ATP EdTech summit.

37. **Hao, J.**, Kyllonen, P. C. (Dec. 2019). Scalable assessment of collaboration, Presentation given to the NAEP assessment development group, Educational Testing Service, NJ

38. **Hao, J.**, Kyllonen, P. C. (Oct. 2019). EPCAL: Towards Scalable Assessments of CPS, Presentation for the Technical Advisory Group for the Collaborative problem Solving (CPS) Project, Educational Testing Service, NJ

39. **Hao, J.**, Kyllonen, P. C. (Oct. 2019). Face to face, Online Collaboration and Technology Status, Presentation for the Technical Advisory Group for the Collaborative problem Solving (CPS) Project, Educational Testing Service, NJ

40. Kyllonen, P. C., **Hao, J.** (Oct., 2019). Estimating an individual's contribution to small group performance, Presentation for the Technical Advisory Group for the Collaborative problem Solving (CPS) Projec, Educational Testing Service, NJ

41. Kyllonen, P. C., **Hao, J.**, and Martin-Raugh, M. (July, 2019). Estimating an individual's contribution to small group performance. Paper given in IMPS 2019 Invited Symposium (P. Halpin, Chair), Recent advances in assessing small group contribution. Santiago, Chile: International Meeting of the Psychometric Society.

42. Li, H., Graesser, A., and **Hao, J.**, (June, 2019), Language and group performance on science inquiry in simulated synchronous collaboration, Presented at CSCL 2019, Lyon, France

43. **Hao, J.**, (April, 2019), Targeting the Data and Modeling Challenges from Next Generation Assessments: A Computational Perspective of Psychometrics, Presented at NCME 2019, Montreal, Canada

44. Adams, D., **Hao, J.**, Deane, P., Wylie, C., Stone, E., Feng, G., (April, 2019), Toward Understanding the Progression of Constructed-Response Fluency Across K–12 Age Groups, Presented at NCME 2019, Montreal, Canada

45. Zhang, M., **Hao, J.**, Deane, P., Li, C., (April, 2019), Measuring Writing Translation Using Keystroke Logs, Presented at NCME 2019, Montreal, Canada

46. Deane, P., **Hao, J.**, Zhang, M., Choi, I., (April, 2019), Identifying Repeated Test Takers Using Keystroke Information from Essay Writing, Presented at NCME 2019, Montreal, Canada

47. **Hao, J.**, and Mislevy, R., (2018, July), glassPy data analytics framework for addressing the practical challenges of process data, Presentation given at International Testing Commission Meeting, Montreal, Canada.

48. **Hao, J.**, (2018, July), Data and modeling challenges from innovative assessments, Presentation given at Statistical methods for innovative testing and learning symposium, IMPS satellite meeting, Columbia University, NYC.

49. **Hao, J.**, (2018, June), Assessing 21st century skills via human-human interactions: task design and psychometric challenges, presentation given at ICSA Applied Statistics Symposium, New Brunswick, NJ.

50. **Hao, J.**, (2018, May), Process data solution for game and simulation based assessments, Presentation given at ETS-College Board workshop, Princeton, NJ

51. **Hao, J.**, von Davier, A., Dicerbo, K., Mislevy, R., and Pak Wong (2018 April), Evidence-centered design and computational psychometrics solution for game/simulation based assessments, training workshop at NCME 2018, NYC, NY

52. **Hao, J.**, Flor, M., Liu, L., von Davier, A., Andrews, J., and Kyllonen, P. (2018, April), CPS-evaluator: an automated assessment o collaborative discourse, NCME 2018, NYC, NY.

53. **Hao, J.**, (2018, April), Assessing collaboration at scale: a technology-driven paradigm shift, Invited talk at the School of Information, University of Michigan, Ann Arbor.

54. **Hao, J.**, (2018, Feb.), Innovations in educational measurements: challenges, strategies and examples, Invited guest lecture, School of Education, Harvard University

55. **Hao, J.**, (2017, Nov), Understanding the writing process for better formative feedback: features, subscores and psychometric considerations, Presentation given to the visiting panel of research, ETS, Princeton, NJ.

56. **Hao, J.**, (2017 Nov.), Assessing collaboration at scale, Statistics department, Columbia University, NYC, NY

57. **Hao, J.**, et al., (2017 June), ETS collaborative science assessment prototype: design, data and automated annotation, CSCL 2017, Philadelphia, PA

58. **Hao, J.**, et al., (2017 June), EPCAL: computer-supported collaboration at scale, workshop organized in the CSCL 2017, Philadelphia, PA.

59. **Hao, J.**, (2017 June), Towards a better collaboration via intelligent facilitation, UM-ARL Expert Panel workshop, AMES, IA

60. Zhang, M., **Hao, J.**, Deane, P., and Li, C. (2017, April). Defining Bursts in Writing Processes. NCME2017, San Antonio, TX

61. **Hao, J.**, (2017 April), CPS-rater: automated sequential annotation of collaborative problem solving skills, NCME 2017, San Antonio, TX.

62. Ou, L., **Hao, J.**, Lee, Y., Liu, L., and von Davier, A., (2017 April), Response-time analysis in simulation-based and collaborative assessments, NCME 2017, San Antonio, TX.

63. Zhu, M., Ou, L., **Hao, J.**, and von Davier, A., (2017 April), Designing simulation studies for collaborative tasks: a psychometric perspective, NCME 2017, San Antonio, TX.

64. Liu, L., **Hao, J.**, von Davier, A., Kyllonen, P., (2017 April), Designing principles for collaborative assessment: aligning assessment with how students learn, NCME 2017, San Antonio, TX.

65. **Hao, J.**, von Davier, A., Dicerbo, K., and Mislevy, R., (2017 April), Evidence-centered design and computational psychometrics solution for game/simulation based assessments, training workshop at NCME 2017, San Antonio, TX

66. **Hao, J.**, (2017 March), CPS-rater: automated sequential annotation for conversations in collaboration, ETS collaborative interaction seminar, Princeton, NJ

67. **Hao, J.**, (2017 May), CPS-rater: automated sequential annotation for conversations in online collaboration, ETS Tuckers and Holland Seminar, Princeton, NJ

68. **Hao, J.**, (2017 June), Assessing collaborative problem-solving, Presentation given to the ETS summer interns, Princeton, NJ

69. **Hao, J.**, (2017 May), Assessing collaborative problem-solving: from Tetralogue to EPCAL, ETS Colloquium, Princeton, NJ

70. **Hao, J.**, and Zhang, M., (2016 Dec.), Analysis of student problem-solving process in computer-based tests, presentation to visitors from SMEEA, Princeton, NJ

71. **Hao, J.**, (2016 Oct.), 21st century skills, game-based assessments, and big data analytics, Invited seminar, Beijing Normal University, Beijing, China.

72. **Hao, J.** (2016 Sept.), EPCAL demo, CRESSTCON 2016, Los Angeles, CA

73. **Hao, J.** (2016 July), Collaborative learning and assessment at scale, Seminar given to ETS summer interns, Princeton, NJ.

74. **Hao, J.** (2016, July), Collaborative Problem Solving Skills versus Collaboration outcomes: Findings from Statistical Analysis and Data Mining. 9th International Conference on Educational Data Mining, Raleigh, NC.

75. **Hao, J.**, Liu, L., and von Davier, A., (2016 June), Collaborative Learning and Assessment at Scale, Presentation to ETS ProEthica program team, ETS, NJ.

76. **Hao, J.** (2016 June), EPCAL: ETS Platform for Collaboration Assessment and Learning, Poster presentation at Collective Intelligence 2016, NYC.

77. **Hao, J.** (2016 May), EPCAL: ETS Platform for Collaboration Assessment and Learning, Presentation given to New Product Development at ETS, Princeton, NJ.

78. **Hao, J.** (2016 May), EPCAL: ETS Platform for Collaboration Assessment and Learning, Presentation given at IBM Watson Research, Yorktown, NY

79. **Hao, J.** (2016 April), Assessing collaborative problem solving skills: challenges, strategies and analytics, invited seminar presentation at University of Michigan, Ann Arbor, MI (the seminar will be rescheduled due to the cancellation of flight because of weather)

80. Liu, L., **Hao, J.**, von Davier, A., and Kyllonen, P., (2016 April), Collaborative problem solving assessment: bring social aspect into science assessment, Presentation was given at NCME 2016, Washington DC

81. **Hao, J.** (2016, Jan.), Data mining in a nutshell, Training seminar given to the Research Assistants at ETS, Princeton, NJ.

82. **Hao, J.** (2015, Dec.), Taming the log file: why and how, Tutorial training given to the Data Analyst group at ETS, Princeton, NJ.

83. **Hao, J.** and Liu, L., (2015, Nov.), CPS Frame: a platform for many things, Presentation was given to the CLO group at ETS, Princeton, NJ.

84. **Hao, J.**, (2015, Oct.), Taming the log files of game/simulation-based assessment, Presentation was given to the IDEAS group round table meeting, Princeton, NJ.

85. **Hao, J.**, (2015, Oct.), Learning, assessment, and the need for standards in the digital environment: log file designing matters, Webinar presentation for the Association of Test Publishers.

86. **Hao, J.**, Liu, L., and von Davier, A., (2015, Sept.), CPS Frame: A Platform for Large-Scale Assessment and Development of Collaborative Problem-Solving Skills, GSC initiative Seminar, Princeton, NJ, USA.

87. **Hao, J.**, and von Davier, A., (2015, July). Collaborative problem solving, Presentation was given at the seminar for Nobel Laureate Dr. James Heckman, Princeton, NJ

88. Zhang, M., **Hao, J.**, Li, C., and Deane P., (2015, July). Classification of writing styles using keystroke logs: a hierarchical vectorization approach, IMPS, Beijing, China

89. Wang, Z., **Hao, J.**, Liu, L., Chen, L., and von Davier, A. (2015, June). Automated classification of collaborative problem-solving skills based on chat messages in a science simulation task. Paper presented at the INGRoup Conference, Pittsburgh, Pennsylvania, USA

90. **Hao, J.**, (2015, July). Collaborative problem solving in a nutshell, Presentation given to the 2015 summer intern at ETS, Princeton, NJ

91. **Hao, J.**, (2015, June). Assessing collaborative problem solving with simulation, Presentation was given at 11th International Conference on Computer Supported Collaborative Learning, Gothenburg, Sweden

92. **Hao, J.**, (2015, June). Taming Log Files from Game and Simulation-Based Assessments: Data Models and Data Analysis Tools, Presentation given FASP Initiative Seminar, Princeton, NJ

93. **Hao, J.**, (2015, May). Data model and glassPy for log files, Presentation given to the process data special interest group, Princeton, NJ

94. Andrews, J. J., Kerr, D., Mislevy, R. J., von Davier, A. A., **Hao, J.**, and Liu, L. (2015, April). Using simulation to explore gender and cultural differences during collaboration, National Council on Measurement in Education, Chicago, IL.

95. von Davier, A., **Hao, J.**, Liu, L. and Kyllonen, P. (2015 April), A Collaborative Science Assessment from a Measurement Perspective, National Council on Measurement in Education, Chicago, IL.

96. **Hao, J.**, Liu, L., von Davier, A., and Kyllonen, P. (2015 April), Assessing collaborative problem solving with simulation task, Presentation was given at NCME 2015, Chicago, IL

97. Liu, L., **Hao, J.**, von Davier, A., and Kyllonen, P. (2015 April), Assessing and understanding collaborative problem-solving processes: task design and scoring issues, Presentation was given at NCME 2015, Chicago, IL

98. **Hao, J.**, (2015, April). Facilitating the Evidence Identification: Data Model and Analysis Toolkit for Log files, Presentation was given to the Visiting Panel on Research, Princeton, NJ

99. Von Davier, A., **Hao, J.**, Liu, L. and Kyllonen, P., (2015, April). Assessment of collaborative problem solving, Presentation was given to the Visiting Panel on Research, Princeton, NJ

100. **Hao, J.** (2015 April), Multiple Facets of an Assessment With Collaborative Problem Solving Tasks, Session chaired at the annual meeting of the National Council on Measurement in Education, Chicago, IL.

101. **Hao, J.**, (2015, February). Data mining in a nutshell, Tutorial presentation given at the ETS data-mining Workshop, Princeton, NJ.

102. **Hao, J.**, (2014, September). Assessing Collaborative Problem Solving Skills with Simulations and Games, Invited seminar talk, Department of Psychology, Fordham University, NYC.

103. **Hao, J.**, (2014, November). Assessing CPS in action: practical considerations and preliminary results, invited talk at the Innovative Assessment of Collaboration, Washington DC.

104. **Hao, J.**, von Davier, A., Liu, L., Kyllonen, P., and Zapata-Rivera, D. (2014, April). Investigating collaborative problem solving with extended trialogue tasks. Paper presented at the annual meeting of the National Council on Measurement in Education, Philadelphia, PA.

105. **Hao, J.**, Shu, Z. and von Davier, A., (2014, July) Assessing Students' Performances from Process Data in Scenario-based Tasks: An Edit Distance Approach, Paper presented at 79th annual meeting of Psychometric Society (IMPS 2014), Madison, WI

106. **Hao, J.**, Shu, Z. and von Davier, A. (2014, August). Analyzing process data from game/simulation based task: an edit distance approach, poster presented at the ASESS 2014 workshop at the KDD 2014 conference, NYC

107. Von Davier, A., **Hao, J.** (2014, April). Considerations on Including Collaborative Problem Solving Tasks in Standardized Assessments, Paper presented at the annual meeting of the National Council on Measurement in Education, Philadelphia, PA, April 2014.

108. **Hao, J.** and Liu, L. (2014, September). Probing collaborative problem solving with Tetralogue. The presentation was given at ETS ACBA TDE initiative research seminar, Princeton, NJ.

109. **Hao, J.** (2014, May). Probing the Collaborative Problem-solving Skills via a Collaborative Volcano Task, presentation given to ETS summer interns, Princeton, NJ.

110. **Hao, J.** (2014, November). Big data vs. right data: implications for educational measurement, Presentation was given at ETS Big Data, Big Promises, Big Problems Series, Princeton, NJ.

111. **Hao, J.** (2014, January), Edit distance as a way to score the process data from the simulation, Presentation given at ETS seminar on NAEP TEL task analysis.

112. Liu, L., von Davier, A., **Hao, J.**, Kyllonen, P., and Zapata-Rivera, D. (2014, April). A tough nut to crack: Measuring collaborative problem solving. Paper presented at the annual meeting of the National Council on Measurement in Education, Philadelphia, PA.

113. Bergner, Y., Shu, Z., **Hao, J.**, Zhu, M. and von Davier, A., (2014, July) Visualizing and Clustering Process Data Sequences from a Simulation-Based Task, Paper presented at 79th annual meeting of Psychometric Society (IMPS 2014), Madison, WI

114. Shu, Z., von Davier, A., Zhu, M., Bergner, Y. and **Hao, J.** (2014, July), Using Markov-IRT to Characterize Process Data, Paper presented at 79th annual meeting of Psychometric Society (IMPS 2014), Madison, WI

115. **Hao, J.** (2013, May). Data mining in a nutshell, Presentation given at the ETS summer intern seminar, Princeton, NJ.

116. **Hao, J.** (2013, April). Dark Energy Survey: Understanding the acceleration of the universe. Invited talk was given at the American Physical Society April Meeting, Denver, CO.

117. **Hao, J.** (2012, June). Dark Energy Survey. Invited talk was given at the 45th Fermilab Users' Meeting, Batavia.

118. **Hao, J.** (2012, December). Image Quality of the Dark Energy Survey Camera. Dark Energy Survey Collaboration Meeting, Univ. of Texas A and M, College Station, Texas.

119. **Hao, J.** (2011, August). Dark Energy Survey Camera. Meeting of the Division of Particles and Fields of the American Physical Society, Brown University, Providence, RI.

120. **Hao, J.** (2011, June). Spatially Weighted and Measurement Error Corrected Gaussian Mixture Model in Cluster Detection. Poster presented at Statistical Challenge in Modern Astronomy V, Penn State.

121. **Hao, J.** (2011, May). Intrinsic Alignment of Galaxy Clusters. Poster presented at American Astronomical Society Annual Meeting, Boston, MA.

122. **Hao, J.** (2011, January). GMBCG Galaxy Cluster Catalog. Talk was given at American Astronomical Society Annual Meeting, Seattle, WA.

123. **Hao, J.** (2009, December). Two image-based flatness measurement for the flatness of the multiCCD focal plane. Talk given at Dark Energy Survey Collaboration Meeting, SLAC National Accelerator Laboratory.

124. **Hao, J.** (2009, August). High redshift galaxy clusters in SDSS Coadd. Talk given at Department of Physics and Astronomy, Northwestern University.

125. **Hao, J.** (2009, June). Galaxy Clusters and Cluster member intrinsic alignment. Talk given at Department of Astronomy and Astrophysics, University of Chicago.

126. **Hao, J.** (2009, March). GMBCG Catalog for SDSS Data Release 7. Talk given at Kavli Institute for Particle Astrophysics, Stanford University.

127. **Hao, J.** (2009, January). Optical Cluster Detection At High Redshift. Talk given at Astrophysics Seminar at Fermi National Accelerator Laboratory.

128. **Hao, J.** (2008, May). Error Corrected Gaussian Mixture Model and its Application in Cluster Finding. Talk given at Cluster Cosmology Workshop, Ohio State University.

129. **Hao, J.** (2007, May). Dark Energy Survey and Cluster Cosmology. Talk given at Dark Energy Workshop, Shanghai United Center for Astrophysics, Shanghai, China

130. **Hao, J.** (2006, December). Galaxy Cluster Finding at High Redshift. Talk given at Dark Energy Survey Chicago Collaboration Meeting, the University of Chicago.

131. **Hao, J.** (2006, June). Cosmological Constraints from Supernovae Data Using Robust Analysis. Poster presented at Statistical Challenge in Modern Astronomy IV, Penn State University.